EXP NO:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

AIM:

To upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files
uploaded = files.upload()
file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)
print("File Loaded Successfully:", file_name)
print("\nFirst 5 rows of the dataset:")
print(df.head())
print("\nMissing values:")
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(
    subset=['Product', 'Quantity', 'Region'],
    inplace=True
)
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
df.dropna(subset=['Date'], inplace=True)
print("\nMissing values after preprocessing:")
print(df.isnull().sum())
print("\nSummary Statistics:")
print(df.describe())
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print("\nProduct-wise Sales and Quantity:")
print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(
    product_summary['Product'],
    product_summary['Sales'],
    color='steelblue'
)
plt.xlabel('Product')
plt.ylabel('Total Sales')
```

```
plt.title('Total Sales by Product')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(
    sales_over_time['Date'],
    sales_over_time['Sales'],
    marker='o',
    color='green'
)
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)
print("\nSales by Region and Product:")
print(pivot_table)
numeric_df = df.select_dtypes(include=np.number)
correlation_matrix = numeric_df.corr()
print("\nCorrelation Matrix:")
print(correlation_matrix)
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()
print("\nFinal Preprocessed Dataset:")
print(df.head())
print("\nFinal Dataset Shape:", df.shape)
print("\nData preprocessing and visualization completed successfully!")
```

OUTPUT:

First 5 rows of the dataset:

	Date	Product	Quantity	Sales	Region
0	2026-01-01	Laptop	2	120000.0	North
1	2026-01-02	Phone	5	75000.0	South
2	2026-01-03	Headphones	8	16000.0	East
3	2026-01-04	Tablet	3	45000.0	West
4	2026-01-05	Laptop	1	60000.0	East

Missing values:

Date	0
Product	1
Quantity	0
Sales	1
Region	1

dtype: int64

Missing values after preprocessing:

Date	0
Product	0
Quantity	0
Sales	0
Region	0

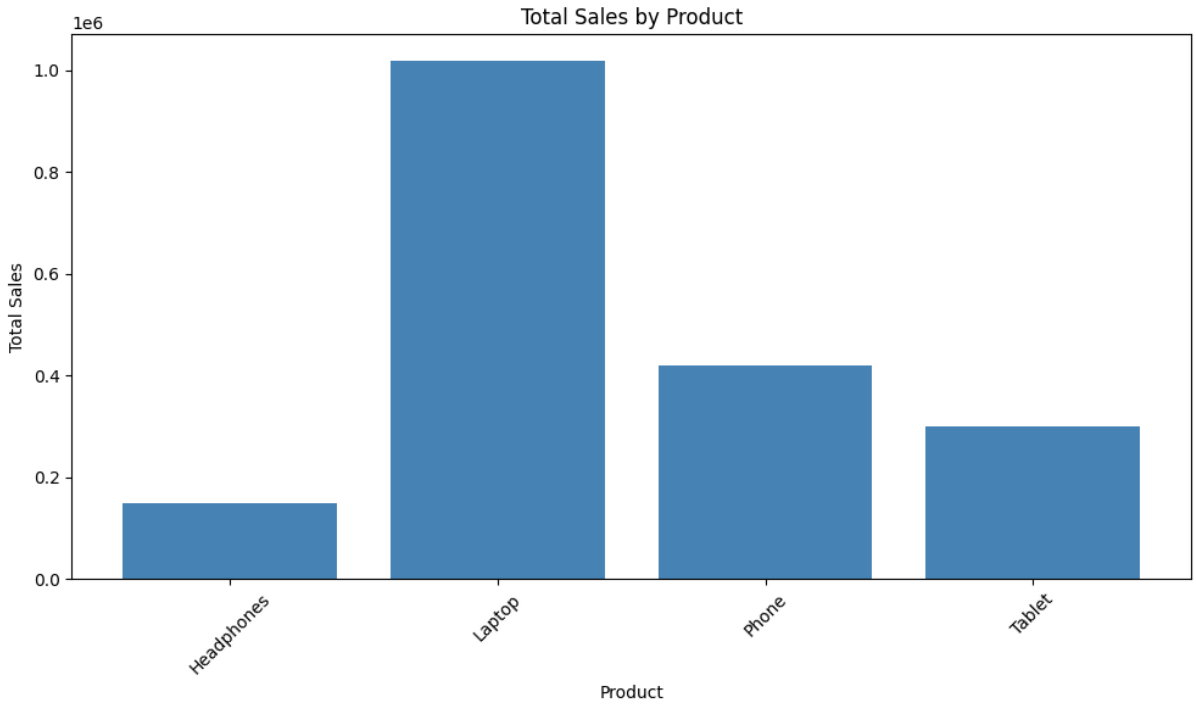
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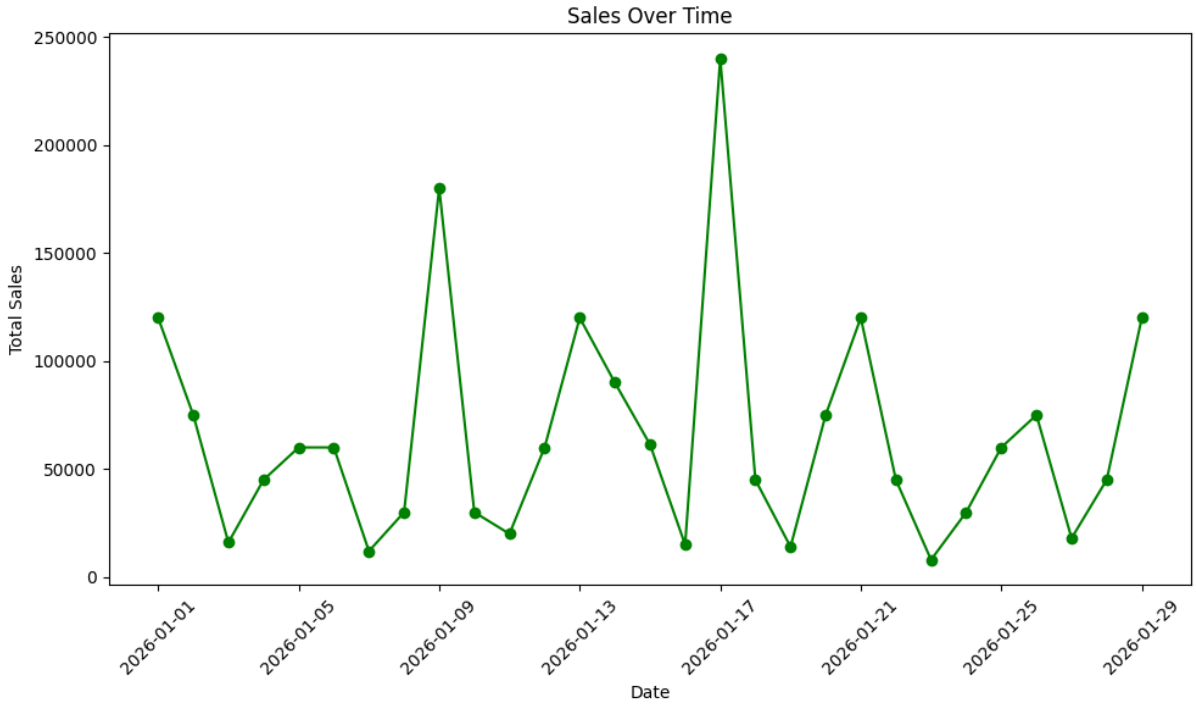
Summary Statistics:

	Date	Quantity	Sales
count	29	29.000000	29.000000
mean	2026-01-15 00:00:00	3.931034	65151.279199
min	2026-01-01 00:00:00	1.000000	8000.000000
25%	2026-01-08 00:00:00	2.000000	30000.000000
50%	2026-01-15 00:00:00	3.000000	60000.000000
75%	2026-01-22 00:00:00	5.000000	75000.000000
max	2026-01-29 00:00:00	10.000000	240000.000000
std	NaN	2.374433	53471.083438

Product-wise Sales and Quantity:

	Product	Sales	Quantity
0	Headphones	1.493871e+05	49
1	Laptop	1.020000e+06	17
2	Phone	4.200000e+05	28
3	Tablet	3.000000e+05	20





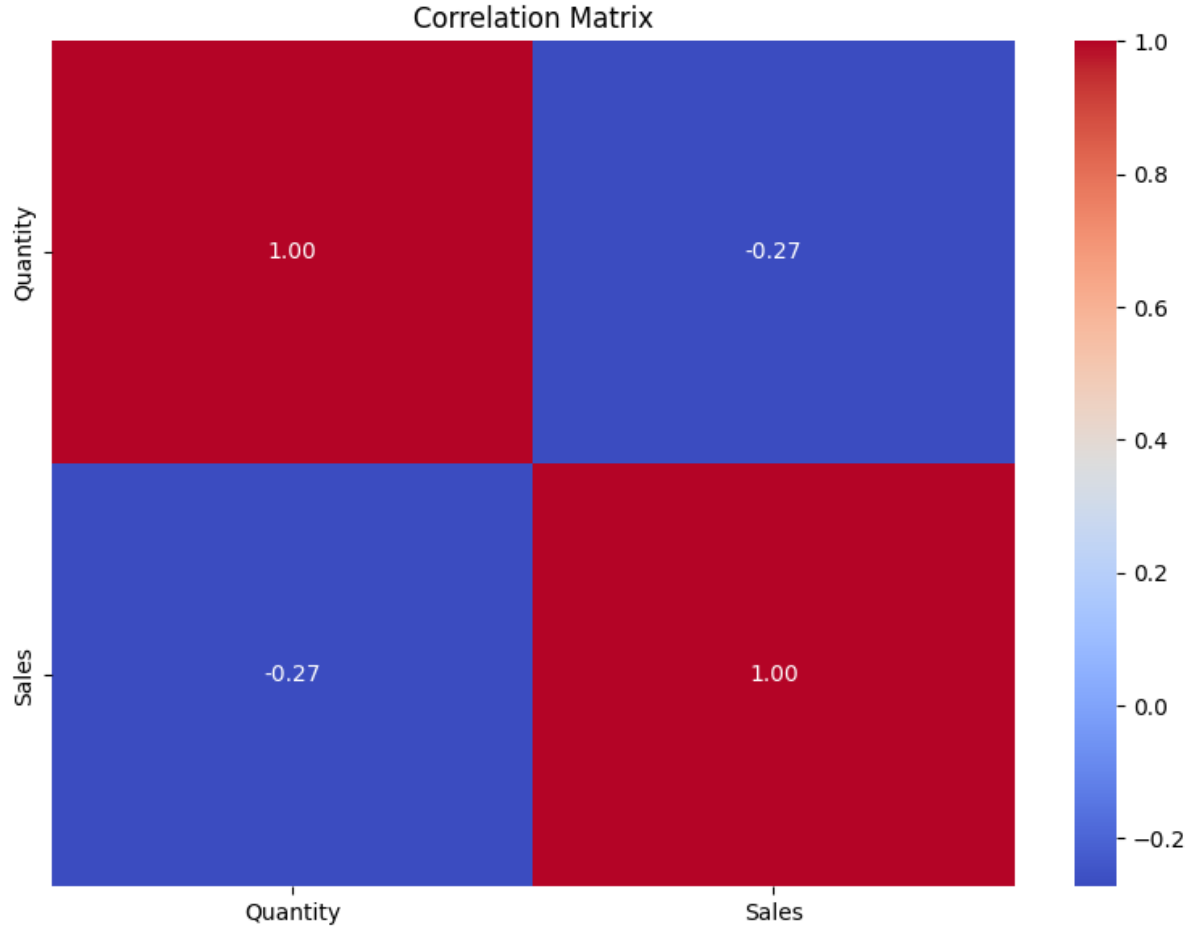
Sales by Region and Product:

Region	Product	Headphones	Laptop	Phone	Tablet
East		30000.000000	120000.0	75000.0	15000.0
North		61387.096774	480000.0	135000.0	30000.0
South		20000.000000	120000.0	120000.0	105000.0
West		38000.000000	300000.0	90000.0	150000.0

Correlation Matrix:

	Quantity	Sales

Quantity 1.000000 -0.272382
Sales   -0.272382 1.000000



Final Preprocessed Dataset:

	Date	Product	Quantity	Sales	Region
0	2026-01-01	Laptop	2	120000.0	North
1	2026-01-02	Phone	5	75000.0	South
2	2026-01-03	Headphones	8	16000.0	East
3	2026-01-04	Tablet	3	45000.0	West
4	2026-01-05	Laptop	1	60000.0	East

Final Dataset Shape: (29, 5)

Data preprocessing and visualization completed successfully!

RESULT:

Thus the program to Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.